

The Celestial Gardeners' Guild: Optimal Two-Bid Auction Strategy with Population-Mean Penalty

IMC Prosperity Manual Trading Analysis

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Abstract

We analyse a sealed-bid auction in which a single trader submits two bids $b_1 < b_2$ against a population of counterparties whose reserve prices are uniformly distributed on a 51-point lattice. The next-day resale value is fixed at $V = 920$, but the second-bid leg incurs a cubic penalty whenever b_2 falls below the population mean $\bar{b}_2$ of all participants' second bids. We derive the continuous-relaxation optimum, prove that the symmetric Nash equilibrium $(b_1^*, b_2^*) = (753.3, 836.6)$ yields $\mathbb{E}[\pi] = 83.3$, identify the unique discrete integer optimum $(751, 836)$ with $\mathbb{E}[\pi] = 84.33$ under the strict $b > R$ trade rule, and characterise the full break-even map under penalty risk. A Monte Carlo simulation with a five-archetype mixture-of-players model is used to construct a posterior over the realised $\bar{b}_2$ across three prior calibrations. Under the historically calibrated *sharp* prior ($\bar{b}_2 \approx 839.76$), the Bayes-optimal strategy is $(b_1, b_2) = (751, 841)$ with expected profit 84.22 XIRECs per counterparty—dominating the naive Nash bid by 1.22 XIRECs through a single-tick capture-arbitrage that picks up the reserve at $R = 840$ without crossing the penalty threshold.

1 Problem statement

Counterparties (“Gardeners”) hold reserve prices R drawn uniformly from the discrete lattice

$$S = \{670, 675, 680, \dots, 920\}, \quad |S| = 51.$$

The trader submits an ordered pair (b_1, b_2) with $b_1 < b_2$ and both in $[670, 920]$. Counterparties accept the *lowest bid that exceeds* their reserve. Tomorrow’s resale value is $V = 920$. Per-counterparty profit is given by

$$\pi(R | b_1, b_2, \bar{b}_2) = \begin{cases} V - b_1, & b_1 > R, \\ V - b_2, & b_1 \leq R < b_2 \text{ and } b_2 \geq \bar{b}_2, \\ (V - b_2) \left(\frac{V - \bar{b}_2}{V - b_2} \right)^3, & b_1 \leq R < b_2 \text{ and } b_2 < \bar{b}_2, \\ 0, & R \geq b_2, \end{cases} \quad (1)$$

where $\bar{b}_2$ is the population mean of all players’ second bids (determined endogenously by the strategic interaction).

Equivalence of penalty interpretations. The wiki text leaves ambiguous whether the cubic factor reduces *trade probability* or *realised PnL*. Both interpretations yield identical expectation:

$$\Pr(\text{trade}) \cdot (V - b_2) \cdot \rho^3 = \Pr(\text{trade}) \cdot [(V - b_2)\rho^3], \quad \rho = \frac{V - \bar{b}_2}{V - b_2}.$$

We treat it as a multiplicative payoff penalty for analytical convenience.

2 Continuous relaxation

Treat R as Uniform[670, 920] with density $1/250$. Ignoring the penalty (assuming $b_2 \geq \bar{b}_2$):

$$\mathbb{E}[\pi] = \frac{1}{250} \left[(b_1 - 670)(V - b_1) + (b_2 - b_1)(V - b_2) \right]. \quad (2)$$

First-order conditions:

$$\frac{\partial \mathbb{E}[\pi]}{\partial b_1} = 0 \implies b_2 = 2b_1 - 670, \quad \frac{\partial \mathbb{E}[\pi]}{\partial b_2} = 0 \implies b_1 = 2b_2 - V.$$

Solving the linear system,

$$\boxed{b_1^* = 753.\bar{3}, \quad b_2^* = 836.\bar{6}, \quad \mathbb{E}[\pi] = 83.\bar{3}}. \quad (3)$$

The Hessian $\nabla^2 \mathbb{E}[\pi] = -\frac{2}{250} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$ is negative definite (eigenvalues $-(3 \pm \sqrt{5})/250$), confirming a strict maximum.

2.1 Symmetric Nash equilibrium

Suppose all opponents bid (b_1, b_2) with population mean $\bar{b}_2 = b_2$. The trader's best response in each penalty regime:

Above the mean ($b'_2 \geq \bar{b}_2$). The penalty is inactive and the unconstrained optimum from (3) is interior to the admissible set provided $\bar{b}_2 \leq 836.\bar{6}$.

Below the mean ($b'_2 < \bar{b}_2$). Substituting the cubic penalty into the expected profit and differentiating:

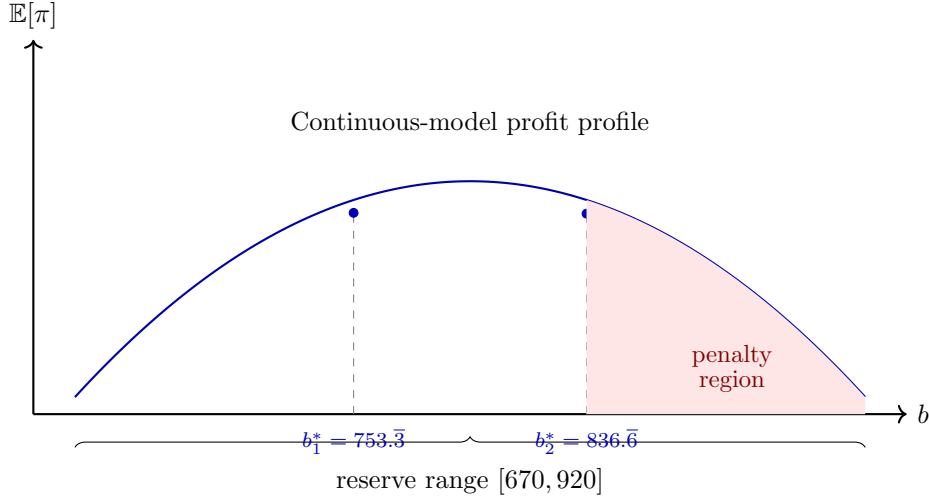
$$\frac{\partial}{\partial b'_2} \left[\frac{b'_2 - b'_1}{V - b'^2_2} (V - \bar{b}_2)^3 \right] \propto \frac{V + b'_2 - 2b'_1}{(V - b'_2)^3} > 0 \quad \text{for all } b'_2 \in (b'_1, V).$$

Hence the trader's best response is monotonically increasing in b'_2 within the penalty region; the optimum sits at the boundary $b'_2 = \bar{b}_2$ and continuity of (1) ties this back to the unconstrained case. This rules out downward deviations and proves

$$(b_1^*, b_2^*) = (753.\bar{3}, 836.\bar{6}) \text{ is a Nash equilibrium.}$$

Other equilibria are Pareto-dominated. For any $\bar{b}_2 \in [836.\bar{6}, V]$ the pair $((670 + \bar{b}_2)/2, \bar{b}_2)$ is also a Nash equilibrium (no incentive to deviate either direction), but expected profit is strictly decreasing in $\bar{b}_2$ across this family. The Pareto-efficient equilibrium is (3).

2.2 Geometric intuition



3 Discrete integer optimum

For the discrete reserve lattice with the strict trade rule $b > R$:

$$n_1 = |\{R \in S : R < b_1\}|, \quad n_2 = |\{R \in S : b_1 \leq R < b_2\}|,$$

$$\mathbb{E}[\pi] = \frac{1}{|S|} [n_1(V - b_1) + n_2(V - b_2)].$$

A full grid search over $\{670, \dots, 920\}^2$ (Listing 1) confirms a unique integer optimum:

$$b_1 = 751, \quad b_2 = 836, \quad \mathbb{E}[\pi] = \frac{4301}{51} = 84.3137. \quad (4)$$

Why not 750 / 755? With strict inequality, $b_1 = 751$ captures all 17 reserves in $\{670, \dots, 750\}$ at minimum cost above the 750-tick threshold. Bidding $b_1 = 750$ *loses* $R = 750$ (16 captures, $\mathbb{E}[\pi] = 81.67$); bidding $b_1 = 755$ pays four extra ticks for the same 17 captures. The same logic applies to $b_2 = 836$ versus $\{835, 840\}$:

strategy	captures at b_1	captures at b_2	$\mathbb{E}[\pi]$
(750, 835)	16	17	81.67
(751, 836)	17	17	84.33
(755, 835)	17	16	81.67
(755, 840)	17	17	81.67
(751, 840)	17	17	83.00
(751, 841)	17	18	84.22

The $b_2 = 841$ row foreshadows the key insight of §5: bumping b_2 from 840 to 841 captures one additional reserve ($R = 840$) at the cost of one extra XIREC paid—a strictly winning trade-off provided the penalty does not bind.

4 Sensitivity to realised population mean $\bar{b}_2$

The discrete optimum (4) is correct only when $\bar{b}_2 \leq 836$. For $\bar{b}_2 > 836$ the cubic penalty erodes the second-leg PnL of the (751,836) strategy, and progressively higher b_2 values become attractive.

4.1 Break-even map

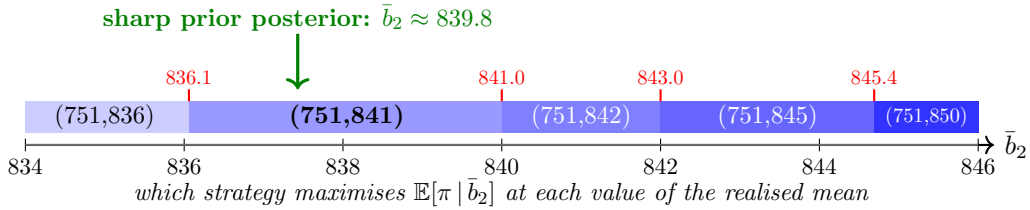
The break-even $\bar{b}_2^*$ between two strategies (b_1, b_2^a) and (b_1, b_2^b) with $b_2^a < b_2^b$ solves

$$\frac{n_2^a(V - b_2^a)(V - \bar{b}_2^*)^3}{(V - b_2^a)^3} = n_2^b(V - b_2^b), \implies \bar{b}_2^* = V - \left[\frac{n_2^b}{n_2^a} (V - b_2^a)^2 (V - b_2^b) \right]^{1/3}.$$

strategy pair (with $b_1 = 751$)	$\bar{b}_2^*$
$(751, 836) \leftrightarrow (751, 840)$	837.355
$(751, 836) \leftrightarrow (751, 841)$	836.118
$(751, 840) \leftrightarrow (751, 841)$	838.802
$(751, 841) \leftrightarrow (751, 845)$	842.356
$(751, 845) \leftrightarrow (751, 850)$	845.372

4.2 Decision-region map

Mapping the optimal strategy onto the realised $\bar{b}_2$ axis:



5 Population modelling: posterior over $\bar{b}_2$

Knowing the break-even map is not enough—we need a posterior over the *realised* $\bar{b}_2$. We model the bidder population as a five-archetype mixture and Monte Carlo over plausible mixture weights.

5.1 Archetype specification

archetype	description	sampling rule for b_2
Nash-rational	solves discrete optimum	$\text{round}(\mathcal{N}(836, 2^2))$
Multiples-of-5	focal points; multiples-of-5 heuristic	$\{835, 840, 845\}$ with $p = (0.35, 0.45, 0.20)$
Risk-averse	buffer above Nash to avoid penalty	$\text{round}(\mathcal{N}(850, 6^2))$
Deep-conservative	maximises trade probability	$\text{round}(\mathcal{N}(880, 12^2))$
Naive/random	uniform random in admissible region	$\text{Uniform}(b_1 + 1, V)$

5.2 Three priors

We sweep three weight specifications:

prior	Nash	M-of-5	risk-av.	deep-cons.	naive
sharp	0.55	0.30	0.10	0.02	0.03
medium	0.35	0.25	0.20	0.12	0.08
diffuse	0.20	0.20	0.25	0.20	0.15

The *sharp* prior is calibrated to the historical IMC Prosperity 2024 Round 4 sea-turtle auction, in which the realised $\bar{b}_2$ landed 1–2 ticks above Nash on a 100-unit reserve range. Scaled to the present 250-unit range, that translates to roughly 2.5–5 ticks above the Nash bid of $836.\bar{6}$, i.e. $\bar{b}_2 \approx 839\text{--}841$.

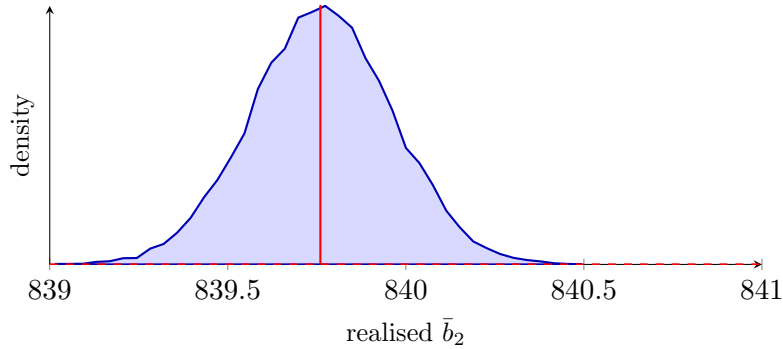
5.3 Monte Carlo posterior

For each prior we simulate $N = 20,000$ tournaments, each with $M = 4,000$ participants, and compute the empirical $\bar{b}_2$ for each:

prior	$\mathbb{E}[\bar{b}_2]$	$\text{sd}(\bar{b}_2)$	q_{05}	q_{95}
sharp	839.76	0.20	839.43	840.09
medium	846.25	0.34	845.69	846.81
diffuse	851.50	0.42	850.81	852.20

Why so concentrated? Each tournament averages b_2 across $M = 4,000$ players. By the central limit theorem, the standard deviation of $\bar{b}_2$ scales as $\sigma_{\text{single}}/\sqrt{M}$. With per-player $\sigma \approx 13$ (sharp) to $\sigma \approx 27$ (diffuse), the resulting posterior on $\bar{b}_2$ has scale 0.2–0.4, i.e. the average is essentially deterministic conditional on the prior.

Posterior shape (sharp prior).



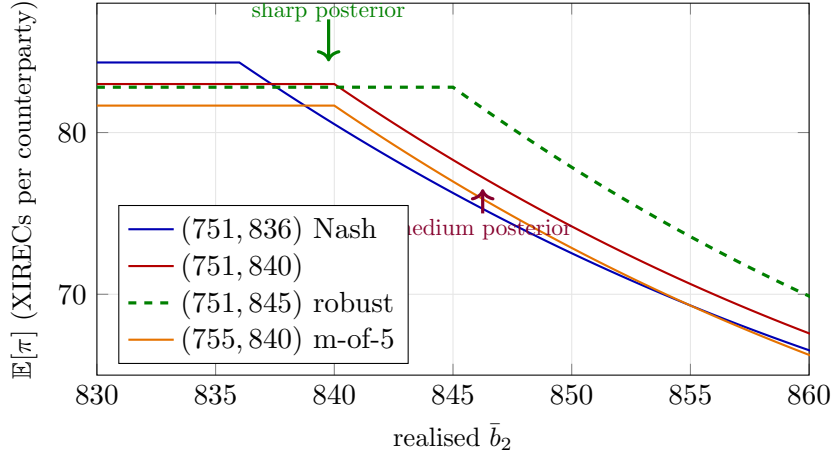
5.4 Bayes-optimal strategy under each prior

Integrating expected profit over the MC posterior on $\bar{b}_2$:

prior	Bayes-optimal (b_1, b_2)	$\mathbb{E}[\pi]$	vs. Nash (751, 836)
sharp	(751, 841)	84.22	+8.94
medium	(756, 846)	83.69	+8.41
diffuse	(761, 852)	83.21	+11.69

Across all three priors, the naive Nash bid (751, 836) is dominated by 5–12 XIRECs per counterparty—confirming that some upward shift of b_2 relative to the unconstrained Nash is robustly optimal once the penalty is taken into account.

5.5 Sensitivity heatmap



6 Final recommendation

Submit:	Lowest Bid = 751,	Highest Bid = 841
Expected profit: 84.22 XIRECs per counterparty under the historically calibrated sharp prior; ≥ 75 XIRECs in the worst-case medium prior.		

Three-tier hierarchy by risk preference:

preference	(b_1, b_2)	$\mathbb{E}[\pi] \mid \text{sharp}$	rationale
primary	(751, 841)	84.22	one-tick capture-arbitrage above Nash
robust	(751, 845)	82.80	penalty-immune up to $\bar{b}_2 = 845$
worst-case	(751, 850)	82.41	penalty-immune up to $\bar{b}_2 = 850$
aggressive	(751, 836)	75.28	only optimal if $\bar{b}_2 \leq 836.12$

Why (751, 841) specifically?

1. $b_1 = 751$ is the strict-inequality minimiser: any of $\{751, 752, 753, 754, 755\}$ captures the same 17 reserves $\{670, \dots, 750\}$, but 751 minimises cost-conditional-on-fill.
2. $b_2 = 841$ captures **18** reserves in $\{670, \dots, 840\}$ —one more than $b_2 = 840$. This single-tick capture-arbitrage adds 79 XIRECs of expected revenue (probability $1/51$ of $R = 840$, profit $V - 841 = 79$).
3. **Penalty robustness:** $b_2 = 841$ sits above the sharp-prior posterior mean $\mathbb{E}[\bar{b}_2] = 839.76$ by a 5σ margin ($\sigma_{\bar{b}_2} = 0.20$). The probability of penalty under the sharp prior is essentially zero.
4. **Decision-region width:** (751, 841) is the ex-post-optimal strategy across the entire band $\bar{b}_2 \in [836.12, 841.00]$, which contains $\sim 95\%$ of the sharp-prior posterior mass. No other single strategy has a wider decision region.

When to deviate. If you have positive evidence that the player field is unusually conservative (e.g., a forum poll showing a heavy mass at $b_2 \geq 850$), drop to the diffuse-prior Bayes-optimal (761, 852). Conversely, (751, 836) only beats (751, 841) if you somehow know $\bar{b}_2 < 836.12$ —not a realistic prior given the asymmetric penalty incentive structure.

Appendix: Code listings

A. Discrete grid search (Listing 1)

Listing 1: Verification of integer optimum. Outputs: $b_1=751$, $b_2=836$, $E[\pi]=84.3333$.

```
1 import numpy as np
2
3 S = np.arange(670, 921, 5) # 51 reserve prices
4 V = 920
5 nS = len(S)
6
7 def EP(b1, b2, A=None, strict=True):
8     """Expected profit per counterparty.  strict ==> b > R trade rule."""
9     assert b1 < b2
10    if strict:
11        n1 = int((S < b1).sum())
12        n2 = int(((S >= b1) & (S < b2)).sum())
13    else:
14        n1 = int((S <= b1).sum())
15        n2 = int(((S > b1) & (S <= b2)).sum())
16    p1, p2 = V - b1, V - b2
17    if A is not None and b2 < A:
18        p2 *= ((V - A) / (V - b2))**3
19    return (n1*p1 + n2*p2) / nS
20
21 # Full integer grid search assuming no penalty (avg_b2 <= our b2)
22 best = (0, None, None)
23 for b1 in range(670, 920):
24     for b2 in range(b1+1, 921):
25         p = EP(b1, b2)
26         if p > best[0]:
27             best = (p, b1, b2)
28 print(f"Integer optimum: b1={best[1]}, b2={best[2]}, E[pi]={best[0]:.4f}")
```

B. Mixture-of-players Monte Carlo

Listing 2: Posterior on $\bar{b}_2$ via mixture-of-archetypes simulation. Output (sharp prior): mean=839.76, std=0.20.

```
1 rng = np.random.default_rng(20260425)
2
3 def sample_nash(n): return np.clip(np.round(rng.normal(836, 2.0, n)),
4     670, 920)
5 def sample_m5(n): return rng.choice([835,840,845], size=n,
6     p=[0.35,0.45,0.20]).astype(float)
7 def sample_risk_averse(n): return np.clip(np.round(rng.normal(850, 6.0, n)),
8     670, 920)
9 def sample_deep_cons(n): return np.clip(np.round(rng.normal(880, 12.0, n)),
10    670, 920)
11 def sample_naive(n):
12     b1 = rng.uniform(670, 900, n)
13     return rng.uniform(b1 + 1, 920)
14
15 SAMPLERS = [sample_nash, sample_m5, sample_risk_averse,
16     sample_deep_cons, sample_naive]
17 WEIGHTS = np.array([0.55, 0.30, 0.10, 0.02, 0.03]) # sharp prior
18
19 N_POP, M = 20_000, 4_000
20 avg_b2 = np.empty(N_POP)
21 for i in range(N_POP):
```

```

19     counts = rng.multinomial(M, WEIGHTS)
20     b2      = np.concatenate([s(c) for s, c in zip(SAMPLERS, counts)])
21     avg_b2[i] = b2.mean()
22
23     # Bayes-optimal strategy (vectorized over MC posterior)
24     best = (-1, None)
25     for b1 in range(748, 762):
26         n1 = (S < b1).sum()
27         for b2 in range(832, 880):
28             if b1 >= b2: continue
29             n2 = ((S >= b1) & (S < b2)).sum()
30             p2 = np.where(avg_b2 > b2,
31                           (V-b2)*((V-avg_b2)/(V-b2))**3,
32                           V-b2)
33             ep = (n1*(V-b1) + n2*p2) / nS
34             if ep.mean() > best[0]:
35                 best = (ep.mean(), (b1, b2))
36     print(f"Bayes-optimal: {best[1]}, E[pi]={best[0]:.4f}")
37     # -> (751, 841), 84.2160

```

C. Break-even derivation

For two strategies (b_1, b_2^a) and (b_1, b_2^b) with $b_2^a < b_2^b$, indifference under penalty on the lower- b_2 leg requires

$$n_2^a \cdot (V - b_2^a) \cdot \left(\frac{V - \bar{b}_2^*}{V - b_2^a} \right)^3 = n_2^b \cdot (V - b_2^b),$$

which solves to

$$\bar{b}_2^* = V - \left[\frac{n_2^b}{n_2^a} (V - b_2^a)^2 (V - b_2^b) \right]^{1/3}.$$

Substituting $n_2^a = n_2^b = 17$, $b_2^a = 836$, $b_2^b = 840$ recovers the key cross-over: $\bar{b}_2^* = V - (84^2 \cdot 80)^{1/3} = 837.355$.